

Profile

Generative AI and Machine Learning engineer with hands-on experience in **LLM fine-tuning**, **Retrieval-Augmented Generation (RAG)**, and **agent-based AI workflows**. Skilled in building **proof-of-concept (POC) AI systems** involving document understanding, decision routing, and autonomous pipelines. Strong Python foundation with exposure to model training, evaluation, and lightweight deployment.

Education

IIIT Sri City	2022 – 2026
B.Tech in Electronics and Communication Engineering – CGPA: 7.79	Sri City, AP

Technical Skills

Programming: Python, C++, JavaScript (basic), SQL
Generative AI & NLP: Transformer Fine-Tuning (DistilBERT), Hugging Face, RAG Pipelines, Embeddings, Prompt Engineering
Agentic AI: Multi-Step Agent Workflows, Decision Routing, Tool-Oriented AI Pipelines
ML Foundations: Classification, Model Evaluation (Precision, Recall, F1, mAP), Hyperparameter Tuning
Frameworks & Tools: PyTorch, Scikit-learn, Transformers, Pandas, NumPy, FAISS, Streamlit, FastAPI (basic), Git

Projects

Retrieval-Augmented Generation (RAG) Document QA System	<i>POC Project</i>
- Built a RAG-based question-answering system over PDF documents using chunking, embeddings, and vector search. - Implemented semantic retrieval with FAISS and lightweight sentence-transformer embeddings to ground LLM responses in source documents. - Reduced hallucinations by enforcing source-backed answer generation with citations, improving response reliability.	
Agentic AI News Intelligence System (LLM Fine-Tuning & Agents)	GitHub
- Designed an end-to-end agent-based AI pipeline for news ingestion, preprocessing, classification, and automated decision-making. - Fine-tuned a pretrained DistilBERT transformer for multi-class news classification, achieving ~80% accuracy. - Implemented confidence-based decision routing agents that triggered alerts, logging, or storage actions based on prediction reliability.	
AI Interview Helper Bot (Real-Time GenAI System)	<i>Full-Stack + GenAI</i> — GitHub
- Built a real-time AI-assisted interview system using WebRTC for low-latency, event-driven interaction. - Designed an LLM-driven question generation and response pipeline with clear session lifecycle and state management. - Integrated OpenAI Speech-to-Text (STT) for live transcription and Grok / LLM APIs for adaptive interview responses. - Implemented Redis-based session caching for active interviews, avoiding repeated MongoDB reads and improving response latency by ~10–15%.	

Experience

AI/ML Research Intern — NSTL (DRDO)	May – July 2025
Visakhapatnam, Andhra Pradesh	
- Developed deep learning pipelines for underwater object detection using YOLOv8, YOLOv11, and Faster R-CNN on 8,000+ sonar images. - Conducted preprocessing, dataset normalization, and performance evaluation using Precision, Recall, F1-score, and mAP.	

Optional Engineering Project

Real-Time Intrusion Detection System	GitHub
- Built a real-time intrusion detection system using YOLOv8 and OpenCV on live video streams. - Automated alerting via email and WhatsApp with snapshots and short video clips, reducing manual monitoring by ~70%.	

Achievements & Certifications

- IBM: Python for AI, ML, and Data Science
- Solved **400+ DSA problems** across LeetCode and GFG